Ananth Raman

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Education

Columbia-Juilliard Program — New York, NY
Intended Majors: Computer Science (Columbia), Trumpet (Juilliard) — *Class of 2030*

Bridgewater-Raritan High School — Bridgewater, NJ
Valedictorian; National Merit Scholarship Winner; NJ Seal of Biliteracy (Spanish) — *Class of 2026*

Research Experience

Advised by Prof. Ambuj Tewari — *University of Michigan, Ann Arbor, MI*

Generation from Noisy Examples | *ICML 2025* — arXiv:2501.04179
Ananth Raman, Vinod Raman

- Formalized generation from noisy data, where an adversary inserts finitely many incorrect examples into the example stream, by defining new noisy notions of generatability that extend prior work on clean data
- Characterized when classes are uniformly generatable under noise using a new dimension, the Noisy Closure dimension, and showed every finite and countable class remains generatable despite finite noise

Apple Tasting: Combinatorial Dimensions and Minimax Rates | *COLT 2024* — arXiv:2310.19064
V. Raman, U. Subedi, A. Raman, A. Tewari

- Studied online binary classification under apple tasting feedback, a partial-feedback setting where the learner observes the true label only when it predicts "1."
- Helped close an open question by showing the Littlestone dimension characterizes agnostic apple tasting, and introduced the Effective width, which tightly characterizes the realizable case and gives a trichotomy of rates

Multiclass Online Learnability under Bandit Feedback | *ALT 2024* — arXiv:2308.04620
A. Raman, V. Raman, U. Subedi, I. Mehalel, A. Tewari

- Studied online multiclass classification under bandit feedback, where the learner only learns whether each prediction was correct, not the true label, even when the label space is unbounded
- Proved that finiteness of the Bandit Littlestone dimension is necessary and sufficient for bandit online learnability, extending prior work on finite label sets

Coursework & Self-Study

Spinning Up in Deep RL (OpenAI) | *Python, PyTorch, Deep Reinforcement Learning* — Self-Study
- Implemented RL algorithms (VPG, TRPO, PPO, DDPG, TD3, SAC) from scratch

Natural Language Processing with Deep Learning (Stanford, CS224N) | *PyTorch, NLP* — Self-Study
Data Structures in Java (COMS W3134) · **Discrete Mathematics** (COMS W3203) — Columbia

Math Programs & Competitions

NSF FinDS: Foundations in Data Science | *UC San Diego* — Summer 2024
Columbia Science Honors Program — Fall 2023
Ross Mathematics Program (declined) · **New York Mathematics Circle** — 2023
Rutgers Young Scholars Program in Discrete Mathematics — Summer 2022
Competitions: AIME Qualifier (4x, 2021–2024) · ARML Qualifier (2025) · NJML Winner (2024, 2025)

Technical Skills

Programming: Python, Java, LaTeX
Machine Learning: PyTorch, Deep Reinforcement Learning, NLP, Online Learning

Music Performance

Columbia University Orchestra · New York Youth Symphony — 2026 – Present
Carnegie Hall's National Youth Orchestra of the USA & NYO2 | *International Tours* — 2024 – 2025
Principal Trumpet: New Jersey Youth Symphony, NAfME All-Eastern, NJMEA All-State — 2022 – 2026
Winner, Princeton Brass Band Youth Solo Competition — Spring 2022